

Machine Learning Lab

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Lab 4

Part 1:

1.

```
library(data.table)
library(ggplot2)
library(mlbench)
library(elasticnet)      # New
library(mclust)          # New
library(scatterplot3d)
```

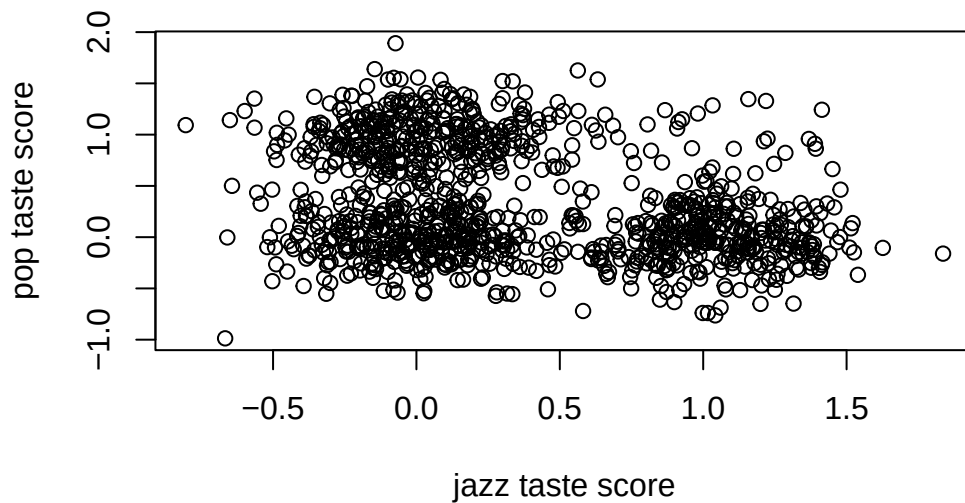
```
taste_influence <- read.csv("~/Documents/ML/Labs/Lab4/taste_influence.csv")

str(taste_influence)
```

```
'data.frame':  1075 obs. of  4 variables:
 $ jazz      : num  -0.0374 0.5575 -0.083 0.1175 0.2406 ...
 $ pop       : num  -0.0284 0.1587 -0.0207 1.3964 1.0173 ...
 $ hiphop    : num   1.1741 -0.0301 -0.1194 -0.067 0.661 ...
 $ influence: num   1.6763 0.0749 -0.0961 0.3659 1.8119 ...
```

```
plot(taste_influence$jazz, taste_influence$pop,
     xlab = "jazz taste score",
     ylab = "pop taste score",
     main = "Scatterplot of jazz vs pop taste")
```

Scatterplot of jazz vs pop taste



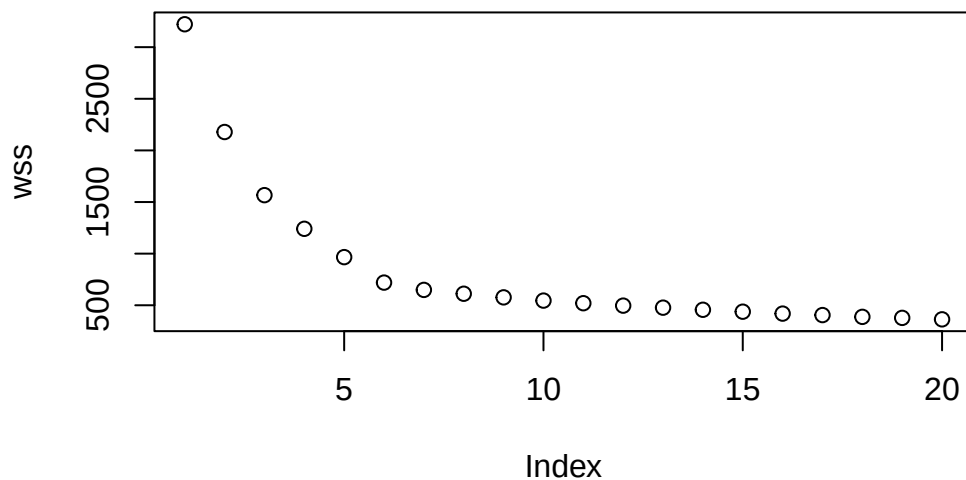
From this two-dimensional perspective, three centers can be identified with respect to jazz and pop taste. One observation is that individuals with a high jazz taste score tend to have a low pop taste score, whereas individuals with a low jazz taste score may have either high or low pop taste scores.

2.

```
taste <- taste_influence[, c("jazz", "pop", "hiphop")] |> scale() |> as.matrix()
```

3.

```
ks <- 1:20
wss <- c()
for(i in 1:length(ks)){
  temp <- kmeans(x = taste, centers = ks[i], nstart = 100) # Use nstart=100 for stability
  wss[i] <- temp$tot.withinss # sum(temp_withinss)
}
plot(wss)
```



According to the elbow method, choosing $k = 6$ is appropriate, since adding further complexity only brings marginal returns.

4.

```
set.seed(42)
finalk <- kmeans(x = taste, centers = 6, nstart = 100)
finalk$centers
```

	jazz	pop	hiphop
1	-0.7075403	-0.6519389	0.9275712
2	-0.5225678	1.2650179	0.8785221
3	1.2617093	-0.6363058	-0.8861036
4	1.1975400	-0.5153682	0.9374598
5	-0.6606565	-0.6264451	-0.8310363
6	-0.6034402	1.2157422	-0.8781105

The centroids indicate that the six clusters capture distinct musical tastes, suggesting that the clusters are meaningfully distinct.

Table 1: Cluster centroids based on musical taste scores

Cluster	Jazz taste score	Pop taste score	Hip-hop taste score	Interpretation
1	Low	Low	High	Hip-hop oriented group
2	Low	High	High	Pop + hip-hop mixed group
3	High	Low	Low	Jazz oriented group
4	High	Low	High	Jazz + hip-hop mixed group
5	Low	Low	Low	Low engagement group
6	Low	High	Low	Pop oriented group

5.

```
set.seed(42)
finalk_test <- kmeans(x = taste, centers = 2, nstart = 100)
finalk_test$centers
```

```
      jazz      pop      hiphop
1 -0.6634616  0.3705600  0.04963173
2  1.1107205 -0.6203654 -0.08308994
```

In the case of $k = 2$:

Cluster 1 can be understood as a pop-oriented group, characterized by relatively low levels of jazz orientation but a stronger inclination toward pop. Hip-hop taste does not play a defining role for this group.

Cluster 2 represents a jazz-oriented group, marked by a pronounced preference for jazz and a weaker inclination toward pop, with hip-hop again contributing little to the differentiation.

Taken together, the two-cluster solution highlights a clear contrast between jazz-oriented and pop-oriented individuals, while hip-hop does not emerge as a key factor in the separation. This suggests that individuals with strong jazz orientation are particularly distinct. Although this simplified specification inevitably loses some of the nuance captured in solutions with more clusters, it remains straightforward and appropriate if the primary interest is in the relationship and distribution between jazz and pop music tastes.

6.

```
taste_influence$cluster <- factor(finalk$cluster)

lm1 <- lm(influence ~ cluster, data = taste_influence)
coef(lm1)
```

```
(Intercept)    cluster2    cluster3    cluster4    cluster5    cluster6
  1.80201909   0.18783818 -0.02645162   0.81590479 -1.60423667 -1.07885163
```

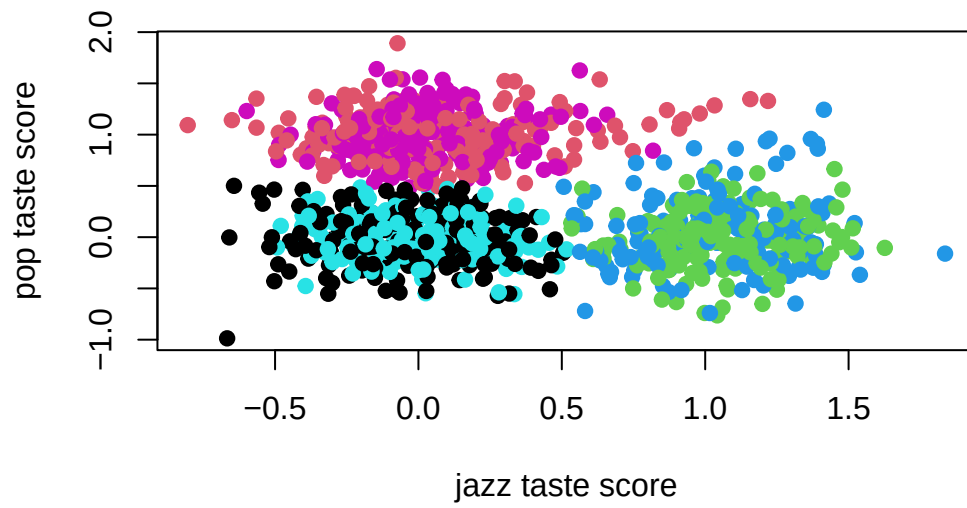
```
# summary(lm1)
```

The linear regression results show that there are significant differences in influence across clusters. The reference cluster is Cluster 1, with an average influence score of 1.802 units. Cluster 4 has the highest influence, about 0.816 units higher than Cluster 1, while Cluster 5 has the lowest influence, about 1.600 units lower than Cluster 1.

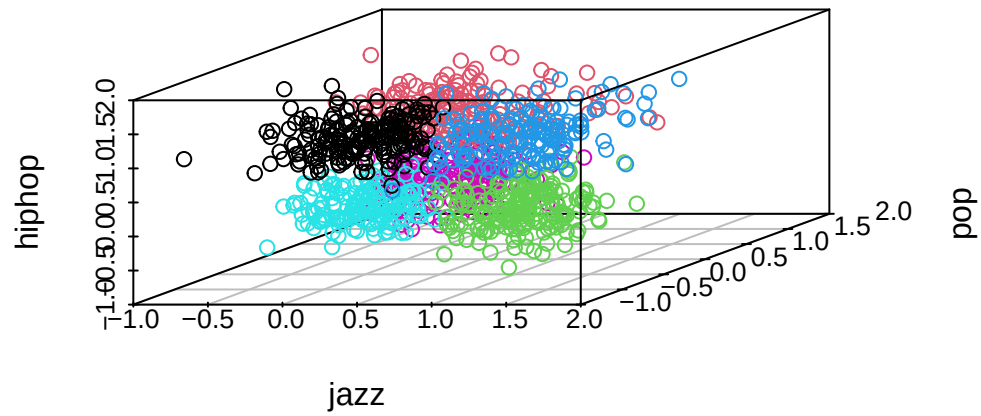
7.

```
plot(taste_influence$jazz, taste_influence$pop,
     xlab = "jazz taste score",
     ylab = "pop taste score",
     main = "Scatterplot of jazz vs pop taste",
     pch = 19, col = taste_influence$cluster)
```

Scatterplot of jazz vs pop taste



```
scatterplot3d(taste_influence[, 1:3], color = taste_influence$cluster)
```

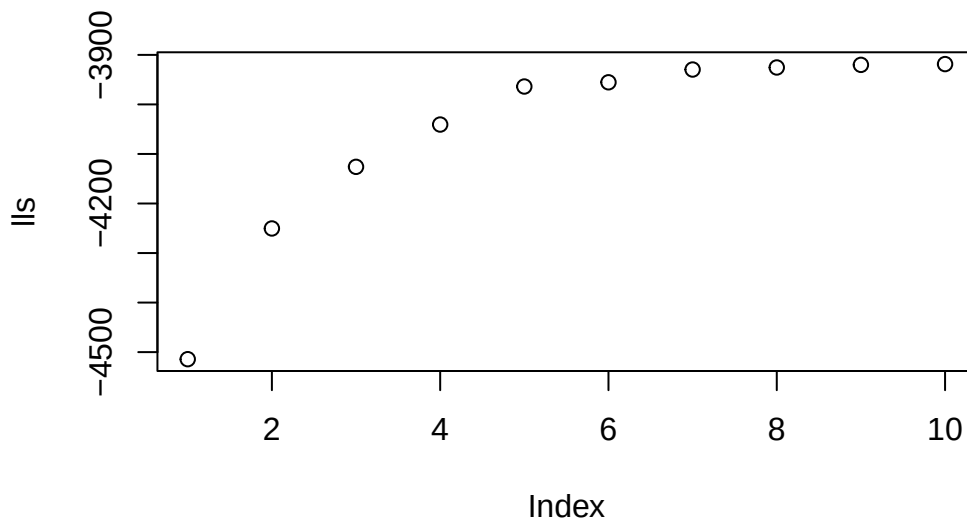


The clustering patterns are captured, and the classification appears clearer. In the two-dimensional jazz-pop view, the six clusters overlap in pairs, which makes them appear as three groups in the earlier plot.

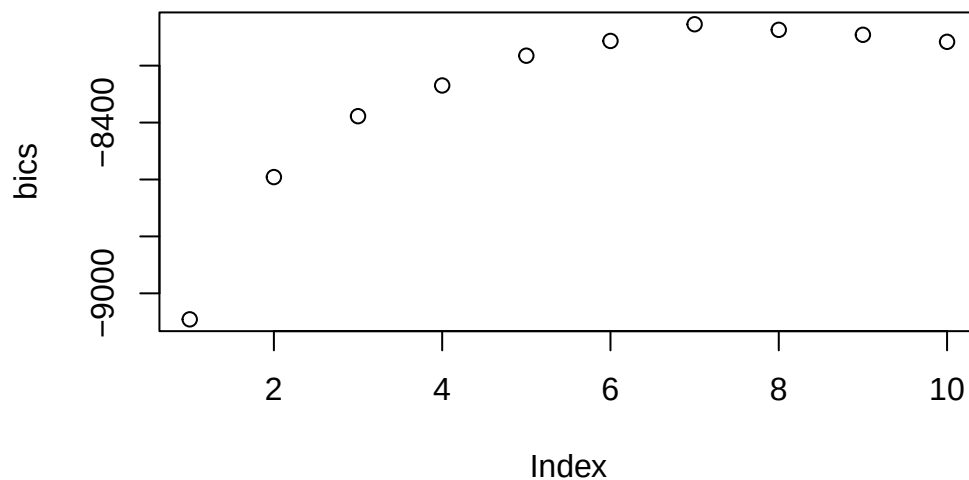
By examining the three-dimensional plot, it can be seen that the clusters are more clearly separated, although some boundaries are still close to each other.

8.

```
lls <- bics <- c()
ks <- 1:10
for(i in 1:10){
  temp <- mclust::Mclust(data = taste, G = ks[i])
  lls[i] <- temp$loglik
  bics[i] <- temp$bic
}
plot(lls)
```



```
plot(bics)
```



According to the elbow method, choosing $G = 7$ is appropriate, since adding further complexity only yields marginal returns.

```
gmm <- mclust::Mclust(data = taste) # If we don't specify "G", mclust will try to find an optimal number of clusters
gmm$G # 7 here (because it uses BIC as criteria)
```

```
[1] 7
```

```
gmm$bic
```

```
[1] -8054.557
```

```
gmm$parameters$mean
```

```
      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
jazz -0.6881655 -0.6493541 -0.5909529 -0.6951585  1.2606871  1.1720341
pop  -0.6460921 -0.6091402  1.2169070  1.2151297 -0.6421940 -0.6766994
hiphop 0.9196229 -0.8299227 -0.8628565  0.8672693 -0.8762022  0.9056929
      [,7]
jazz  1.1080265
pop   1.1541214
hiphop 0.9649712
```

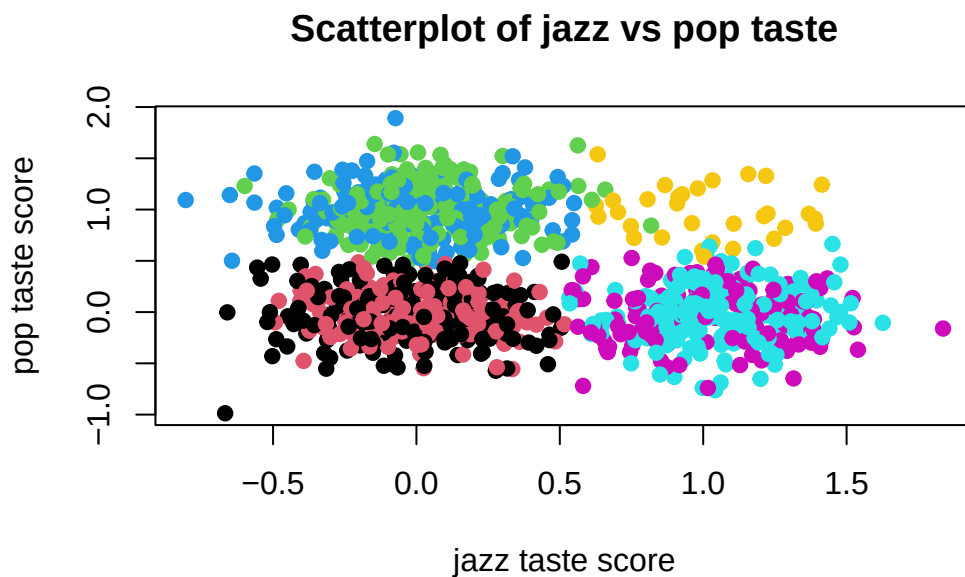
```
taste_influence$cluster_gmm <- factor(gmm$classification)

lm2 <- lm(influence ~ cluster_gmm, data = taste_influence)
coef(lm2)
```

```
(Intercept) cluster_gmm2 cluster_gmm3 cluster_gmm4 cluster_gmm5 cluster_gmm6
1.79641713 -1.59863471 -1.08408350 0.13308067 -0.03081441 0.78398173
cluster_gmm7
1.28185389
```

```
# summary(lm2)

plot(taste_influence$jazz, taste_influence$pop,
     xlab = "jazz taste score",
     ylab = "pop taste score",
     main = "Scatterplot of jazz vs pop taste",
     pch = 19, col = taste_influence$cluster_gmm)
```



```
scatterplot3d(taste_influence[, 1:3], color = taste_influence$cluster_gmm)
```

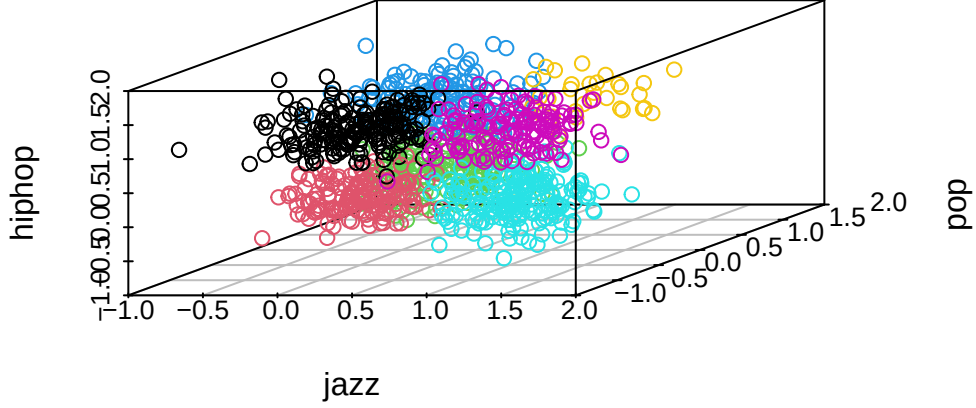


Table 2: Cluster centroids based on musical taste scores (GMM, $G = 7$)

Cluster	Jazz taste score	Pop taste score	Hip-hop taste score	Interpretation
1	Low	Low	High	Hip-hop oriented group
2	Low	Low	Low	Low engagement group
3	Low	High	Low	Pop oriented group
4	Low	High	High	Pop + hip-hop mixed group
5	High	Low	Low	Jazz oriented group
6	High	Low	High	Jazz + hip-hop mixed group
7	High	High	High	All-round music lovers

The distinction appears meaningful, as the clusters represent substantively different groups.

The linear regression results indicate significant differences in influence across clusters. Cluster 1 serves as the reference group, with an average influence score of 1.796 units. Cluster 7 exhibits the highest influence, about 1.281 units higher than Cluster 1, whereas Cluster 2 shows the lowest influence, about 1.599 units lower than Cluster 1.

The clustering patterns are well captured, and the classification becomes clearer. In the two-dimensional jazz–pop view, the seven clusters tend to overlap in pairs, which makes them appear as three broader groups in the earlier visualization.

By examining the three-dimensional plot, it is evident that the clusters are more distinctly separated, although some of the boundaries remain close to each other.

9.

```
taste_influence$uncertainty <- gmm$uncertainty
lm2 <- lm(influence ~ cluster_gmm + uncertainty, data = taste_influence)
coef(lm2)
```

```
(Intercept) cluster_gmm2 cluster_gmm3 cluster_gmm4 cluster_gmm5 cluster_gmm6
 2.06168116 -1.61357422 -1.07142415  0.24290959 -0.09050299  0.81508107
cluster_gmm7 uncertainty
 1.43921977 -4.42903960
```

```
# summary(lm2)
```

The linear regression results with uncertainty included show that there are significant differences in influence across clusters. Cluster 1 serves as the reference group, with an average influence score of 2.062 units. Cluster 7 exhibits the highest influence, about 1.439 units higher than Cluster 1, whereas Cluster 2 shows the lowest influence, about 1.614 units lower than Cluster 1.

The coefficient for uncertainty is -4.43 , indicating that when the uncertainty of a cluster assignment increases by 1, the average influence decreases by approximately 4.43 units.

In other words, individuals who are more “ambiguous” (i.e., located near the cluster boundaries) tend to have lower influence.

Part 2

1.

```
neighborhood <- read.csv("~/Documents/ML/Labs/Lab4/neighborhood.csv")
dim(neighborhood)
```

```
[1] 200 25
```

```
colnames(neighborhood)
```

```

[1] "taste_jazz"           "taste_classical"
[3] "taste_blues"          "taste_pop"
[5] "taste_country"        "taste_raegge"
[7] "income"               "nbhood_avg_income"
[9] "education"            "nbhood_avg_education"
[11] "nhood_crime"           "nbhood_unemployment"
[13] "nhbood_avg_temp"      "nhbood_pop"
[15] "nhbood_nr_lights"     "nhbood_nr_pizzerias"
[17] "city_avg_taste_jazz"  "city_avg_taste_classical"
[19] "city_avg_taste_blues" "city_avg_taste_pop"
[21] "city_avg_taste_country" "city_avg_taste_raegge"
[23] "taste_film_action"    "taste_film_romcom"
[25] "taste_film_documentary"

```

The dataset neighborhood.csv consists of 200 observations and 25 numeric variables. These variables can be grouped into four main categories: 1. Individual taste preferences, such as preferences for music genres and film types; 2. Neighborhood-level characteristics, including crime, unemployment, population size, and average temperature; 3. City-level average taste measures, reflecting aggregated cultural preferences at the city level; 4. Socioeconomic indicators, such as income and education.

2.

```

pca1 <- prcomp(x = neighborhood)
pca1$rotation[, 1:2]

```

	PC1	PC2
taste_jazz	-1.130734e-02	-0.0023440600
taste_classical	-7.785547e-03	0.0075890578
taste_blues	-1.094852e-02	0.0036306694
taste_pop	-7.826432e-05	-0.0116550537
taste_country	1.995948e-03	-0.0017553217
taste_raegge	4.944613e-03	-0.0147713147
income	5.135923e-03	0.0159201632
nbhood_avg_income	6.926408e-03	0.0164944063
education	-2.588437e-03	0.0090498047
nbhood_avg_education	-5.864163e-03	0.0133028161
nhood_crime	4.772406e-03	-0.0105344496
nbhood_unemployment	4.138436e-03	-0.0096372982
nhbood_avg_temp	4.802633e-02	0.5671282890

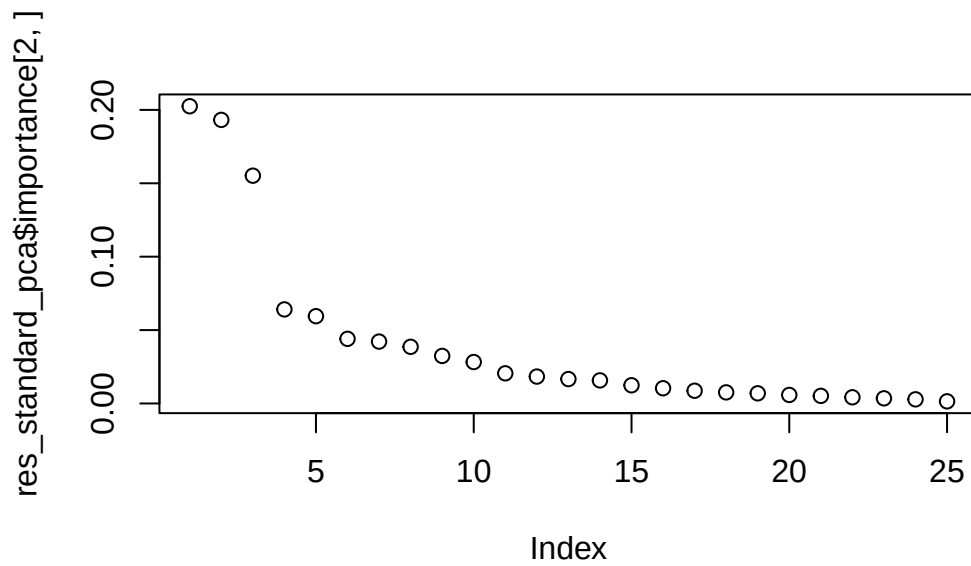
nhbood_pop	-2.325264e-01	0.4612561449
nhbood_nr_lights	-9.704813e-01	-0.1031315060
nhbood_nr_pizzerias	-2.950091e-02	0.6727586139
city_avg_taste_jazz	-5.433612e-03	-0.0054625216
city_avg_taste_classical	7.555348e-03	-0.0079776471
city_avg_taste_blues	3.312960e-03	-0.0006816765
city_avg_taste_pop	-6.397509e-03	0.0016809645
city_avg_taste_country	3.592366e-03	0.0052711518
city_avg_taste_raegge	1.429978e-03	0.0080628332
taste_film_action	1.278440e-02	-0.0064266489
taste_film_romcom	-8.944392e-03	-0.0154741603
taste_film_documentary	-5.620100e-03	0.0218192137

Without standardization, the loadings are not on the same scale, since the unscaled variables are measured in different magnitudes. This makes the loadings incomparable, so the data must be scaled first.

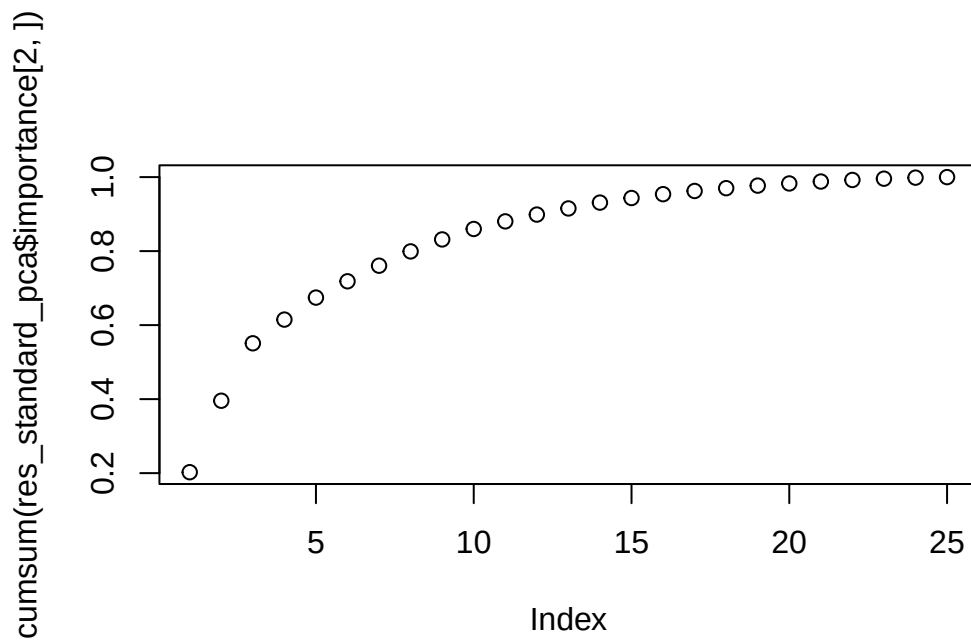
3.

```
neighborhood_scaled <- scale(neighborhood)
pca2 <- prcomp(neighborhood_scaled)
```

```
res_standard_pca <- summary(pca2)
# res_standard_pca$importance
plot(res_standard_pca$importance[2,])
```



```
plot(cumsum(res_standard_pca$importance[2,]))
```



According to the proportion of variance explained plot, we can retain 3 PCs to explain about

55% of the variance, or 10 PCs to explain around 80%. Here we choose 3 PCs for easier interpretation while still capturing about half of the variance.

4.

```
# Principal component loadings
pca2$rotation[,c(1:3)]
```

	PC1	PC2	PC3
taste_jazz	-0.165598284	-0.315796907	0.188815999
taste_classical	-0.186439330	-0.282337949	0.220597843
taste_blues	-0.197560893	-0.315298024	0.166961965
taste_pop	0.218843741	0.270293966	-0.188021321
taste_country	0.205061458	0.277200903	-0.206469793
taste_raegge	0.205283637	0.279349334	-0.207769006
income	-0.334078396	0.119288960	-0.200122814
nbhood_avg_income	-0.340921831	0.100533268	-0.182486414
education	-0.341538161	0.082650349	-0.207921612
nbhood_avg_education	-0.349351488	0.090315358	-0.180684902
nhood_crime	0.329488500	-0.103288082	0.217348454
nbhood_unemployment	0.346934465	-0.113247576	0.191109908
nhbood_avg_temp	0.033748193	0.010148644	-0.045986155
nhbood_pop	-0.036709545	-0.046398844	-0.014724121
nhbood_nr_lights	-0.006280977	-0.004203786	0.018981095
nhbood_nr_pizzerias	-0.110745654	0.026056839	0.061859046
city_avg_taste_jazz	0.095354373	-0.243906113	-0.308085582
city_avg_taste_classical	0.067175435	-0.280305639	-0.289200006
city_avg_taste_blues	0.097818272	-0.278059681	-0.271576796
city_avg_taste_pop	-0.076748148	0.263499641	0.319545930
city_avg_taste_country	-0.091673374	0.247513332	0.295280303
city_avg_taste_raegge	-0.112170433	0.248201800	0.290919907
taste_film_action	0.020026975	-0.063442131	0.002728314
taste_film_romcom	0.049294821	0.113532618	-0.006786904
taste_film_documentary	-0.059787432	-0.053353936	0.029209916

The principal components involve too many variables, making it difficult to provide easy and substantively meaningful interpretations.

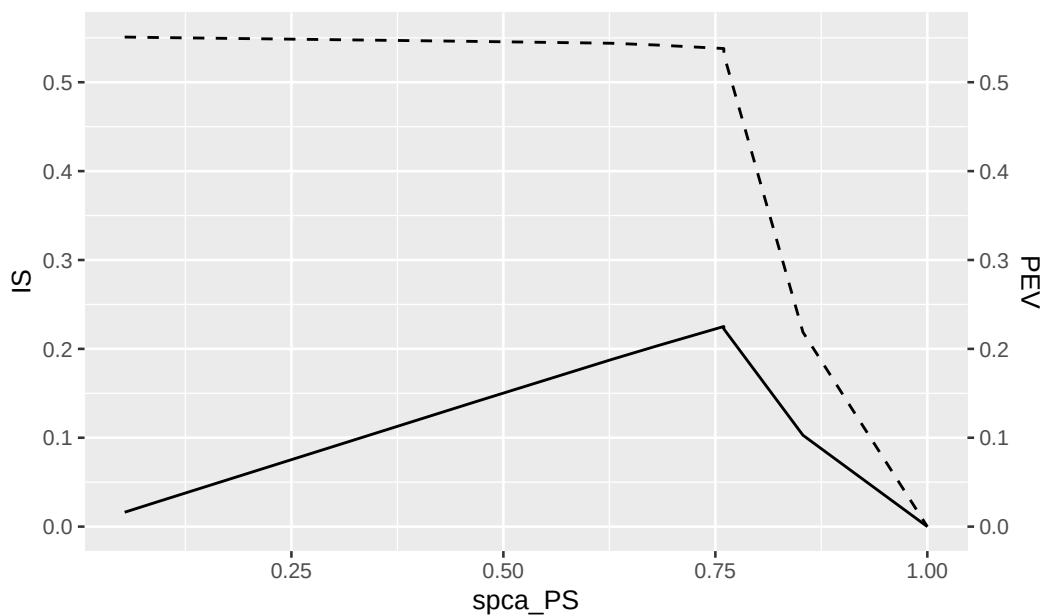
5.

```
# Sparse PCA comparison
penalties <- c(0,1,5,10,20,80,100,200,500,1000)
spca_pevs <- c()
spca_ps <- c()
for(i in 1:length(penalties)){
  temp <- spca(x = neighborhood_scaled,
               K = 3,
               type = 'predictor',
               para = c(rep(penalties[i],ncol(neighborhood_scaled))),
               sparse = 'penalty')
  spca_pevs[i] <- sum(temp$pev)
  spca_loadings <- temp$loadings
  ps <- length(spca_loadings[abs(spca_loadings)<=0.01]) / length(spca_loadings)
  spca_ps[i] <- ps
  if(length(ps)==0){ps <- 0}
}

is_pev_dt <- data.table(lambda=penalties,
                        spca_PEV=spca_pevs,
                        spca_PS=spca_ps)
standard_PCA_PEV <- is_pev_dt[lambda==0]$spca_PEV
is_pev_dt[,IS:=standard_PCA_PEV * spca_PEV * spca_PS]

my_plot <- ggplot(is_pev_dt, aes(x=spca_PS)) +
  geom_line( aes(y=IS)) +
  geom_line( aes(y=spca_PEV),linetype='dashed') +
  scale_y_continuous(name = "IS",
                     sec.axis = sec_axis(~.*1, name="PEV",breaks = scales::pretty_breaks(n=8)),
                     breaks = scales::pretty_breaks(n=8)) +
  ggtitle('PEV x IS x PS for USarrest data') +
  theme_gray(base_size = 10)
plot(my_plot)
```

PEV x IS x PS for USarrest data



```
# Re-fit best spec
final_gmm <- spca(x = neighborhood_scaled,
                  K = 3,
                  type = 'predictor',
                  para = rep(80, ncol(neighborhood_scaled)),
                  sparse = 'penalty')
# final_gmm$loadings
L <- as.data.frame(final_gmm$loadings)
L_nz <- subset(L, PC1 != 0 | PC2 != 0 | PC3 != 0)
L_nz
```

	PC1	PC2	PC3
taste_jazz	0.0000000	-0.4245547	0.0000000
taste_classical	0.0000000	-0.3783390	0.0000000
taste_blues	0.0000000	-0.4234639	0.0000000
taste_pop	0.0000000	0.3953632	0.0000000
taste_country	0.0000000	0.4079272	0.0000000
taste_raegge	0.0000000	0.4178211	0.0000000
income	-0.4257313	0.0000000	0.0000000
nbhood_avg_income	-0.3822598	0.0000000	0.0000000
education	-0.4049781	0.0000000	0.0000000
nbhood_avg_education	-0.3942016	0.0000000	0.0000000

nhood_crime	0.3847728	0.0000000	0.0000000
nbhood_unemployment	0.4529658	0.0000000	0.0000000
city_avg_taste_jazz	0.0000000	0.0000000	-0.3951797
city_avg_taste_classical	0.0000000	0.0000000	-0.4048828
city_avg_taste_blues	0.0000000	0.0000000	-0.4125593
city_avg_taste_pop	0.0000000	0.0000000	0.4568680
city_avg_taste_country	0.0000000	0.0000000	0.3856339
city_avg_taste_raegge	0.0000000	0.0000000	0.3901998

After applying sparse PCA, three principal components can be clearly interpreted.

PC1 loads on income, education, nbhood_avg_income, and nbhood_avg_education (all with negative signs), as well as positively on nhood_crime and nbhood_unemployment. This component distinguishes high-income, well-educated individuals and neighborhoods from low-income, low-education contexts characterized by higher crime and unemployment. It therefore represents a socioeconomic status axis.

PC2 loads on music taste variables, with negative weights for taste_jazz, taste_classical, and taste_blues, and positive weights for taste_pop, taste_country, and taste_raegge. This dimension captures the contrast between “highbrow/traditional musical tastes” (jazz, classical, blues) and “popular/mainstream musical tastes” (pop, country, reggae). Hence, PC2 reflects a musical preference axis.

PC3 loads on city-level average taste variables, with negative weights for city_avg_taste_jazz, city_avg_taste_classical, and city_avg_taste_blues, and positive weights for city_avg_taste_pop, city_avg_taste_country, and city_avg_taste_raegge. This component differentiates cultural environments where cities are oriented toward highbrow music from those where popular music dominates. PC3 can thus be interpreted as a city-level cultural taste axis.

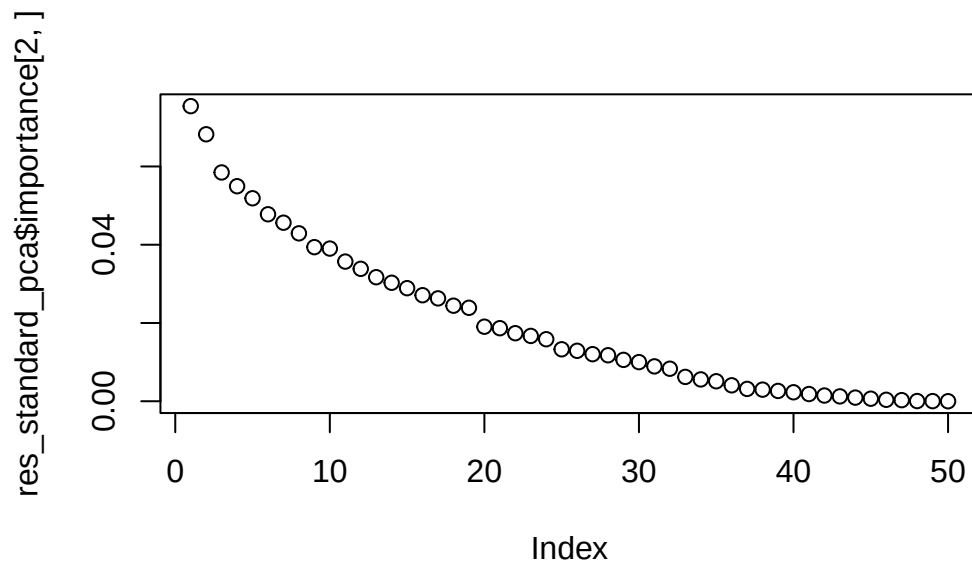
Compared to standard PCA, sparse PCA produces components that are easier to interpret, since each dimension is associated with only a small subset of variables. However, there exists a variance-sparsity trade-off, and sparse PCA sacrifices some explained variance.

6.

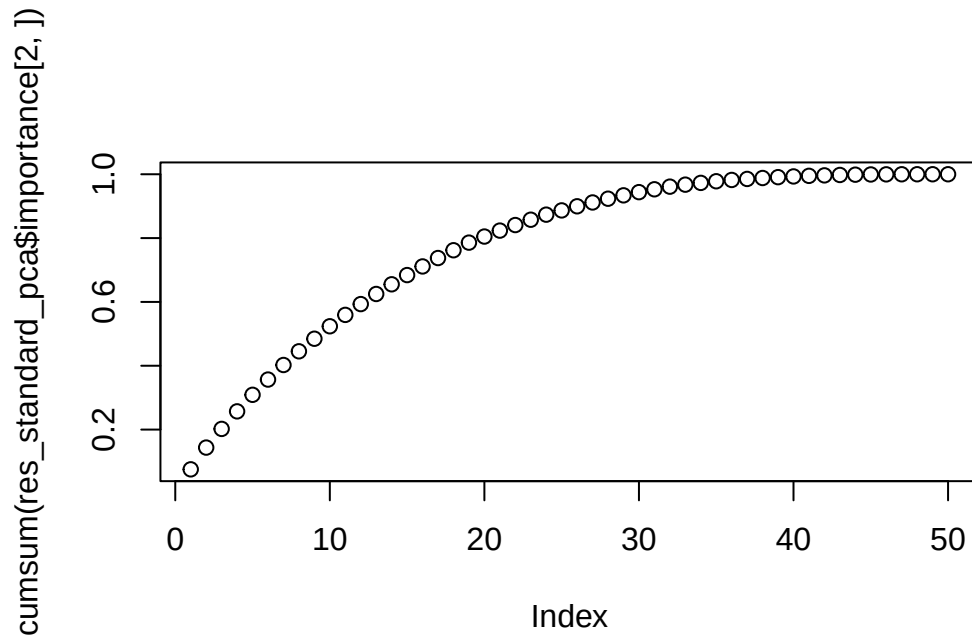
```
gen_data <- function(n,p){
  df <- c()
  for(i in 1:p){
    ith_var <- rnorm(n = n, mean = 0, sd = 1)
    df <- cbind(df,ith_var)
  }
  return(df)
}
```

```
}  
df <- gen_data(50, 50)
```

```
df_scaled <- scale(df)  
pca_df <- prcomp(df_scaled)  
res_standard_pca <- summary(pca_df)  
plot(res_standard_pca$importance[2,])
```



```
plot(cumsum(res_standard_pca$importance[2,]))
```



According to the cumulative variance plot, it is clear that PCA does not perform well, as there is no obvious elbow. In other words, it cannot effectively reduce dimensionality.

The reason is that this dataset does not contain an underlying low-dimensional structure.